Windows Server Basic & Installation

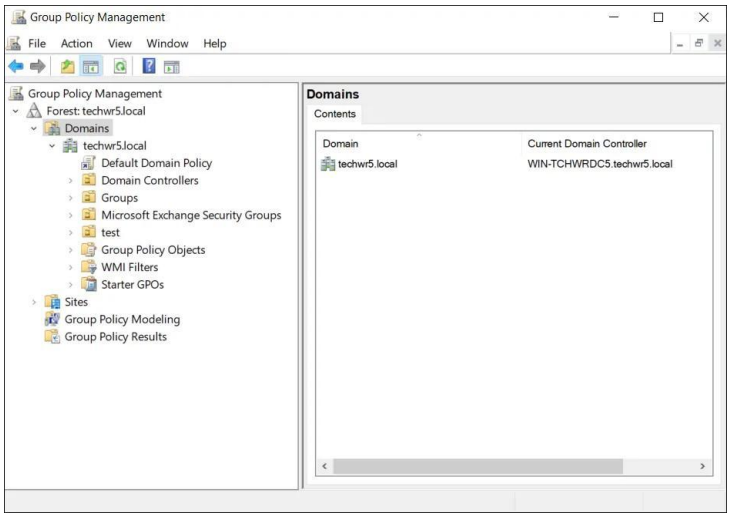
Group Policies Theory :What is Group Policy? Configuring Group Policy Objects (GPOs) to enforce security and administrative settings. Practical :Create and apply GPOs for password policies and desktop restrictions. Task: Document the GPO settings applied and their impact.

1. Open Group Policy Management Console (GPMC) on your Windows Server and create a new GPO named 'PasswordPolicyDemo' that enforces a minimum password length of 8 characters. Link it to your test OU and verify the setting is applied using gpresult.

ANS:

Step 1: Open Group Policy Management

1. Log in to your Windows Server 2019.
2. Open Server Manager.
3. Click Tools in the top-right corner.
4. Select Group Policy Management.

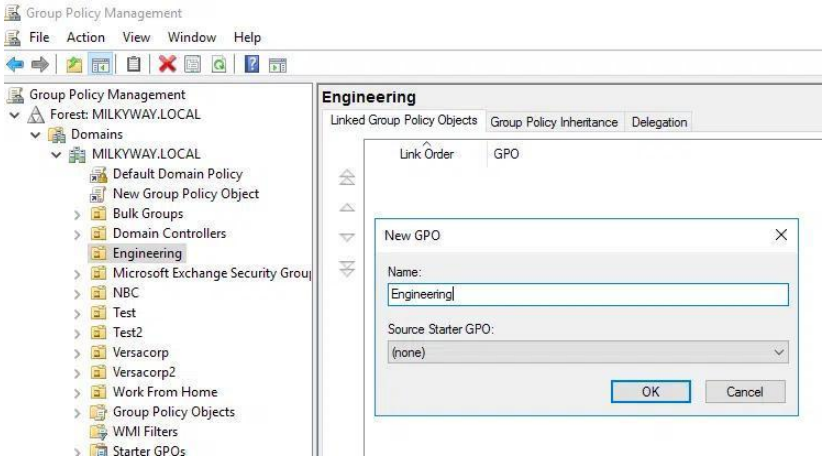


Step 2: Create a new GPO

1. Expand Forest.
2. Expand Domains.
3. Select your domain name.
4. Right-click your domain or the appropriate location for creating the GPO.
5. Select Create a GPO in this domain, and Link it here.

6. Enter the GPO name : PasswordPolicyDemo

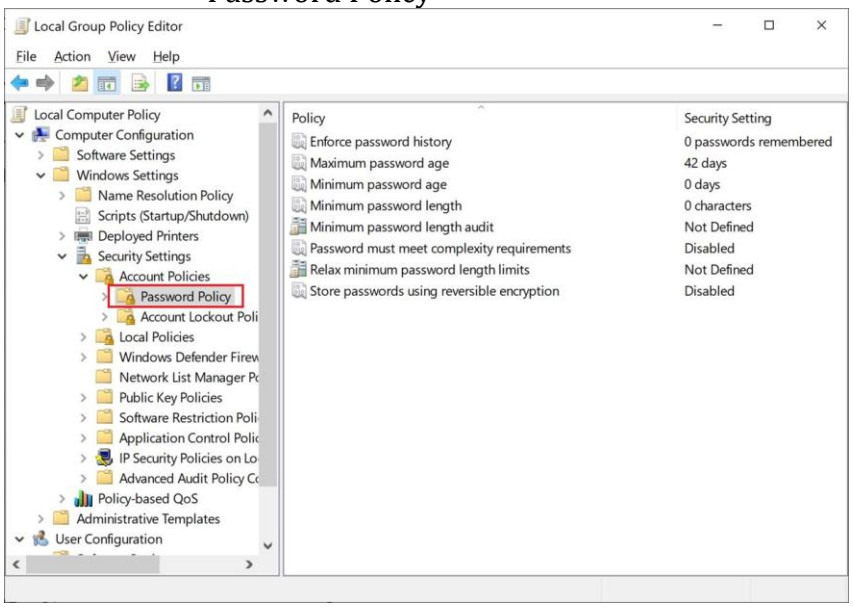
7. Click OK.



Step 3: Edit the PasswordPolicyDemo GPO

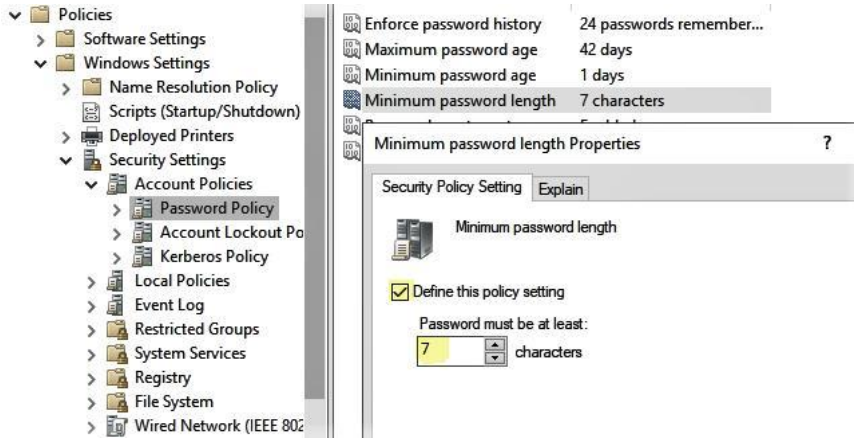
1. Right-click PasswordPolicyDemo.
2. Select Edit.
3. Navigate to:

Computer Configuration
└── Policies
└── Windows Settings
└── Security Settings
└── Account Policies
└── Password Policy



Step 4: Configure minimum password length

1. Double-click Minimum password length.
2. Select Define this policy setting.
3. Enter: 8 characters
4. Click Apply.
5. Click OK.



Step 5: Update Group Policy on the client PC

Log in to your Windows 10 client PC and open Command Prompt as Administrator.

Run: gpupdate /force

2. Configure a GPO to restrict access to the Control Panel for all users in a specific OU, similar to how public computers in a cybercafe or college lab prevent users from changing system settings. Hint: In the GPO, navigate to User Configuration > Administrative Templates > Control Panel and set 'Prohibit access to Control Panel and PC settings' to Enabled.

ANS:

Step 1: Open Group Policy Management

1. Open Server Manager.
2. Click Tools.
3. Select Group Policy Management.
4. Expand your domain.
5. Locate your test OU.

Step 2: Create a new GPO

1. Right-click your test OU.
2. Select Create a GPO in this domain, and Link it here.
3. Enter the GPO name: ControlPanelRestriction
4. Click OK.

Step 3: Edit the GPO

1. Right-click ControlPanelRestriction.
2. Select Edit.
3. Navigate to:

User Configuration
└── Policies
└── Administrative Templates
└── Control Panel

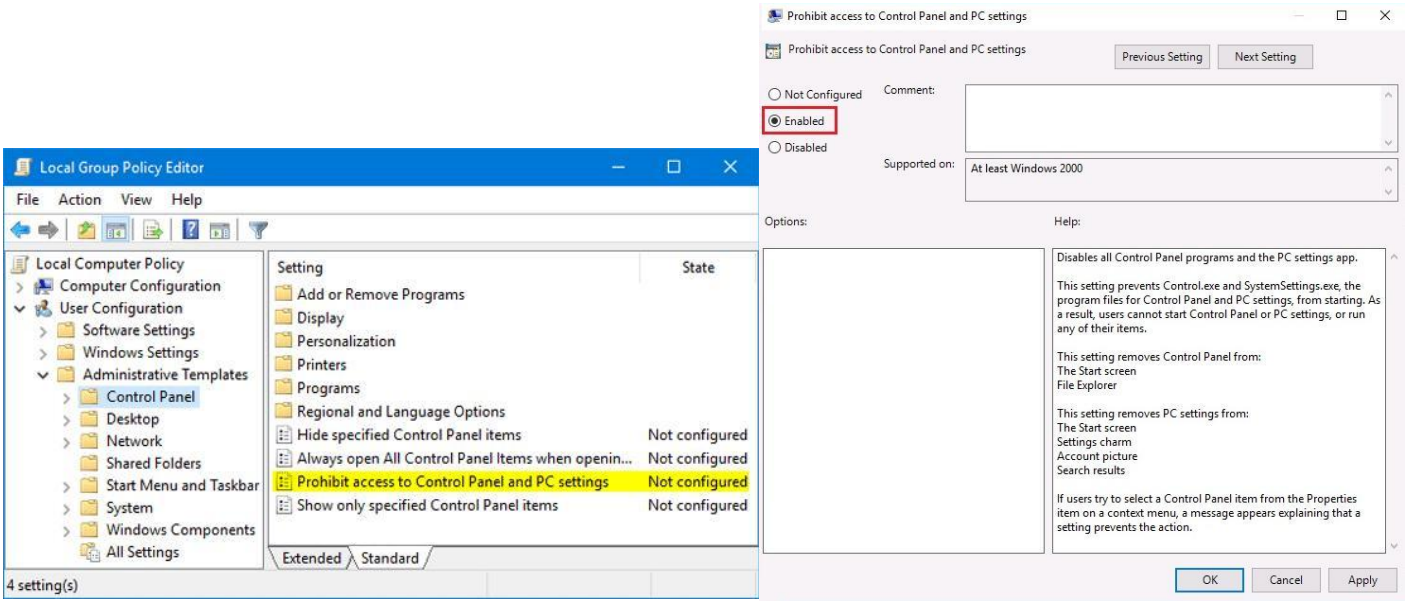
Step 4: Enable the restriction

1. Locate Prohibit access to Control Panel and PC settings.
2. Double-click the policy.
3. Select Enabled.
4. Click Apply.
5. Click OK.

Step 5: Update Group Policy on the client

Log in to a domain user account belonging to the test OU.

Open Command Prompt and run: `gpupdate /force`



3. Edit your 'Password Policy Demo' GPO to enable account lockout after 3 invalid login attempts and set the lockout duration to 10 minutes. Run gpupdate /force on a client PC and test the lockout by entering the wrong password.

ANS:

Step 1: Open PasswordPolicyDemo

1. Open Group Policy Management.
2. Locate PasswordPolicyDemo.
3. Right-click it.
4. Select Edit.

Step 2: Navigate to Account Lockout Policy

Go to:

Computer Configuration
└── Policies
└── Windows Settings
└── Security Settings
└── Account Policies
└── Account Lockout Policy

Step 3: Configure Account Lockout Threshold

1. Double-click Account lockout threshold.
2. Select Define this policy setting.
3. Enter:

3 invalid logon attempts

4. Click Apply.
5. Click OK.

Step 4: Configure Account Lockout Duration

1. Double-click Account lockout duration.
2. Select Define this policy setting.
3. Enter:

10 minutes

4. Click Apply.
5. Click OK.

Step 5: Configure Reset Account Lockout Counter

1. Double-click Reset account lockout counter after.
2. Select Define this policy setting.
3. Enter:

10 minutes

4. Click Apply.
5. Click OK.

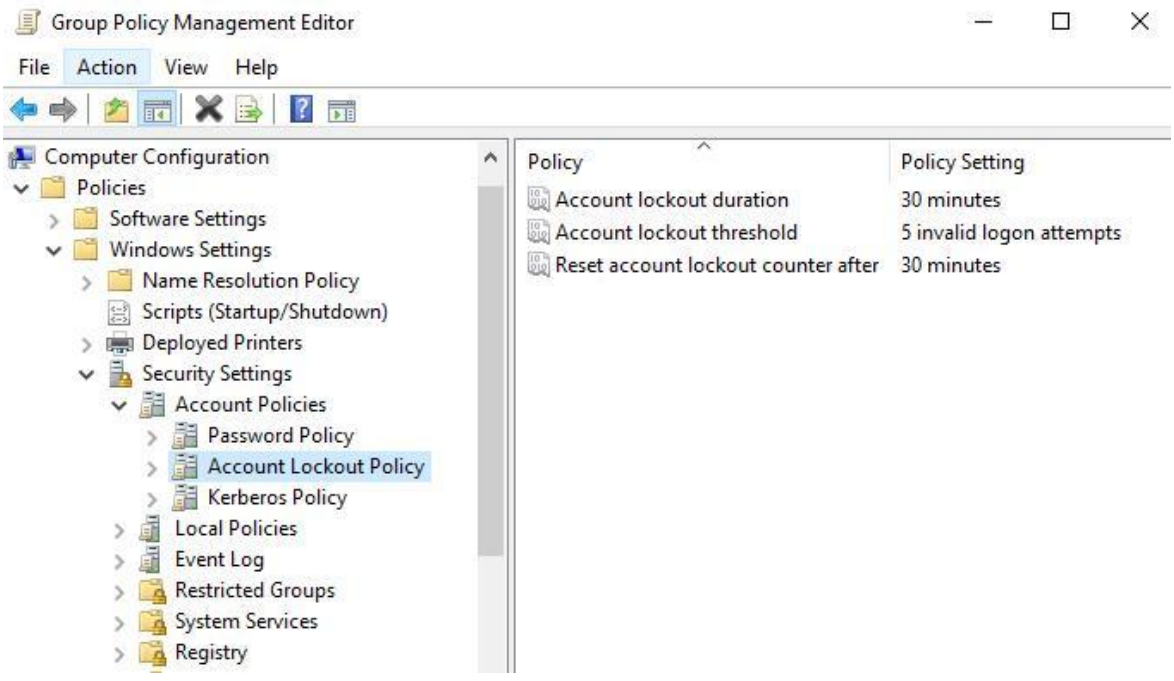
Final Account Lockout Configuration

Policy setting	Configured value
Account lockout threshold	3 invalid attempts
Account lockout duration	10 minutes
Reset account lockout counter after	10 minutes

Step 6: Update Group Policy on the client PC

On your Windows 10 domain-joined client PC, open Command Prompt as Administrator.

Execute: gpupdate /force



4. Document all GPO settings you applied in this session (password policy, desktop restriction, lockout policy) in a table format, and for each setting, describe its expected impact on users in 1-2 sentences.

ANS:

GPO Configuration Summary

Sr. No.	GPO Name	Policy Setting	Configured Value	Expected Impact on Users
1	PasswordPolicyDemo	Minimum password length	8 characters	Users must create passwords containing at least 8 characters. Shorter passwords will not be accepted when setting or changing domain passwords.
2	ControlPanelRestriction	Prohibit access to Control Panel and PC Settings	Enabled	Users in the targeted OU cannot open Control Panel or Windows Settings to modify system configurations.
3	PasswordPolicyDemo	Account lockout threshold	3 attempts	User accounts are locked after three consecutive invalid login attempts, helping prevent unauthorized password guessing.
4	PasswordPolicyDemo	Account lockout duration	10 minutes	Locked accounts automatically become available again after 10 minutes, unless an administrator unlocks them earlier.
5	PasswordPolicyDemo	Reset account lockout counter after	10 minutes	Failed login attempts are reset after 10 minutes without another failed attempt, preventing old failed attempts from accumulating indefinitely.

Explanation of Expected Impact

1. Minimum Password Length

The minimum password length policy improves domain security by requiring users to create passwords containing at least eight characters. It reduces the use of extremely short passwords.

2. Control Panel Restriction

This policy prevents users from accessing Control Panel and PC Settings. It is useful in computer laboratories, colleges, schools, and cybercafes where users should not modify system settings.

3. Account Lockout Threshold

The account lockout threshold protects user accounts against repeated incorrect password attempts. After three consecutive invalid attempts, the account becomes locked.

4. Account Lockout Duration

The account lockout duration determines how long a user account remains locked. After 10 minutes, the account automatically becomes available again.

5. Reset Account Lockout Counter

This setting resets the failed login attempt counter after 10 minutes. It allows users to recover from occasional typing mistakes without permanently accumulating failed attempts.